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ORIGINAL ARTICLE

Instructional Video and GenAI-Supported Chatbot in Digital Game-Based Learning: Influences on Science Learning, Cognitive Load and Game Behaviours

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Keywords: AI in education | chatbots | cognitive load | game-based learning | generative artificial intelligence | inquiry learning | science education

ABSTRACT

Background: Digital game-based learning (DGBL) has shown promise in enhancing learning and motivation, with appropriate scaffolding playing a crucial role in facilitating student inquiries and knowledge acquisition through science games. While scaffolding is generally effective in promoting learning in DGBL, there is variability among different scaffold types, and the impact of scaffolding on cognitive load remains unclear.

Objectives: This study investigates the effects of two scaffolding tools to support secondary students' science learning in a DGBL environment: instructional videos that provide systematic and structured assistance to elucidate underlying science concepts within the game, and a chatbot providing adaptive, contextualised advice and feedback, supported by Generative Artificial Intelligence (GenAI).

Methods: A total of 160 seventh-grade students participated in the study. Using a 2×2 experimental design and a sequential analysis of game trace data, we explore the individual and combined effects of instructional videos and the GenAI-supported chatbot on science learning, game performance, cognitive load and game behaviours.

Results: Results indicate that instructional videos had a significant positive effect on science learning outcomes, with the two groups that had access to instructional videos significantly outperforming the other two groups without the videos, F (1, 153) = 55.64, $p < 0.001$, partial $\eta^2 = 0.27$. Additionally, a significant interaction effect was observed on extraneous cognitive load, with the lowest extraneous load reported by participants who had access to both instructional videos and the chatbot, F (1, 153) = 6.75, $p = 0.01$, partial $\eta^2 = 0.04$. Sequential analysis of game trace data revealed distinct behaviour patterns among the four treatment groups, with the group that had access to GenAI only displaying the most fragmented game inquiry behaviours.

Conclusions: This study contributes to the limited research on the use of instructional videos and GenAI chatbots to scaffold science learning in DGBL environments. The findings highlight instructional videos as a strong scaffold in DGBL. Furthermore, the combination of instructional videos and the GenAI chatbot can result in less extraneous load than the GenAI chatbot alone, while fostering more cohesive inquiry behaviours. Additionally, we underscore the significance of learners' self-regulated learning skills in fully leveraging GenAI in constructivist learning environments like DGBL.

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Summary

- What is already known about this topic
  - Students need scaffolding to acquire science knowledge in DGBL
  - Video-based learning enhances learner engagement, motivation, and performance
  - AI-powered chatbots have the potential to provide adaptive scaffolding.
- What this paper adds
  - Instructional videos provide effective scaffolding in DGBL
  - GenAI-supported chatbot does not show a main effect in supporting science learning in DGBL
  - The combination of instructional videos and GenAI chatbot decreases extraneous cognitive load
  - The combination of instructional videos and GenAI chatbot prompts more cohesive game behaviours
- Implications for practice and/or policy
  - Use instructional videos in DGBL to deliver just-in-time, systematic knowledge
  - Mere access to GenAI support in DGBL does not automatically enhance learning
  - Integrate instructional videos and GenAI chatbot to support learning
  - Prioritise self-regulated learning skills for K–12 students to benefit from GenAI in science inquiries

1 | Introduction

Digital game-based learning (DGBL) refers to the use of digital games designed for educational purposes, aiming to engage learners in meaningful learning through gameplay. These environments combine the motivational appeal of games with targeted educational content, creating interactive experiences that actively promote learning (Park et al. 2019; Prensky 2003; Van Eck and Dempsey 2002). Among various DGBL environments, adventure and simulation games are frequently used in science education (Cheng et al. 2015), where they facilitate inquiry-based learning through immersive adventures and simulations. However, realising DGBL's full potential, particularly in secondary science education, often requires effective scaffolding to help students derive meaningful knowledge from their gameplay (Barzilai and Blau 2014; Cai et al. 2022; Sitzmann 2011; Wouters et al. 2013; Wouters and van Oostendorp 2013).

In Taiwan, as in many other countries, secondary education is a critical component of the K–12 system where STEM subjects, such as physics, become distinct parts of the general science curriculum. However, students entering secondary education often struggle with physics, prompting the adoption of various instructional approaches, such as DGBL, to enhance engagement and promote understanding (Low et al. 2023; Tomara et al. 2017). This study examines an adventure- and simulation-based DGBL environment for secondary physics education, focusing on two scaffolding tools: instructional videos and a chatbot. The instructional videos provide on-demand, structured guidance on the physics concepts embedded in the game. Despite their widespread use, little research has empirically examined their effects in DGBL (Liao et al. 2019). The second scaffold is a chatbot powered by Generative AI (GenAI), offering real time, adaptive, natural-language support during gameplay. Traditionally, chatbots rely on predefined algorithmic or keyword matching to provide support. While these chatbots have shown promise for adaptive scaffolding (Wollny et al. 2021; Wu and Yu 2024), GenAI differs in its ability to generate new content by learning patterns from existing data, enabling chatbots to produce more context-aware, dynamic and human-like responses that adapt to user inputs in real time (Mogavi et al. 2024). Although promising, empirical research on GenAI's effectiveness in DGBL remains limited. Furthermore, it is unclear how GenAI-supported chatbots might complement instructional videos to support science learning among secondary students, who often face motivational challenges (Cai et al. 2022). To address these research gaps, we conducted a 2 × 2 experimental study to investigate the individual and combined effects of these scaffolds in DGBL. Specifically, the study addresses the following research question: In a DGBL environment, does the scaffold of instructional videos, the scaffold of a GenAI-supported chatbot or the interaction between the two impact students' acquisition of science knowledge, cognitive load, in-game performance and game behaviours? In the next section, we provide an overview of the literature that informed the study.

2 | Literature Review

2.1 | DGBL and Scaffolding

DGBL broadly refers to a digital environment where games or game elements are used as motivational drivers to enhance learning (Park et al. 2019; Prensky 2003; Van Eck and Dempsey 2002). Through interactions with game elements such as rules, goals, feedback, conflict and representation (Ke 2009), players achieve learning objectives deliberately embedded within the gameplay (Wouters et al. 2013). DGBL holds potential not only for promoting knowledge acquisition but also for enhancing learner motivation (Connolly et al. 2012; Lamb et al. 2018; Ryan and Rigby 2020; Vlachopoulous and Makri 2017). However, empirical findings reveal both promising (Clark et al. 2016a, 2016b; Sitzmann 2011; Vogel et al. 2006; Wouters et al. 2013) and mixed outcomes, especially in mathematics and science education (Girard et al. 2013; National Research Council 2011; Young et al. 2012).

Different game genres, such as puzzles, action, adventure, strategy, role-playing, simulation and construction, are utilised to support learning (Ke 2016). In science education, adventure and simulation games are the most frequently adopted (Cheng et al. 2015). As learners immerse themselves in-game–world adventures, they interact with simulation elements and engage in inquiry-based processes. Specifically, players manipulate game elements (e.g., adjusting a speed metre) to observe outcomes (e.g., determining whether an object reaches a target within a set time), reflect on cause-and-effect relationships, and make adjustments to achieve goals and earn rewards. This process helps learners uncover scientific principles embedded within the game, fostering deeper conceptual understanding (Ke 2016).

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In this study, the DGBL environment incorporated both adventure and simulation elements, with game mechanics designed to support conceptual learning in physics (Ke 2009; Wouters et al. 2013).

Learners engaging in science inquiries within DGBL often require scaffolding to make sense of their experiences and acquire knowledge (Hmelo-Silver et al. 2007; Kirschner et al. 2006). Scaffolding is especially important in DGBL to help learners connect the game world with abstract science concepts (Barzilai and Blau 2014; Sitzmann 2011; Wouters et al. 2013; Wouters and van Oostendorp 2013). Scaffolding refers to the guidance and support provided by more knowledgeable others, helping learners accomplish tasks that would otherwise be beyond their abilities (Wood et al. 1976). Saye and Brush (2002) categorise scaffolds as *hard* (fixed and static) or *soft* (adaptive and dynamic). In DGBL, scaffolding can take various forms, such as reflections, feedback, hints, prompts, exposition and collaboration (Cai et al. 2022). For example, a study by Chen et al. (2020) found that students playing a science game with question prompts to guide inquiry showed better conceptual understanding and performance than those without prompts. However, Hsu and Tsai (2013) found that students prompted to identify the cause of their failures in a science game did not outperform those without such scaffolds. A recent meta-analysis by Cai et al. (2022) found that scaffolding generally improved learning in DGBL, though with significant variability across studies. Notably, larger effect sizes were observed among college and elementary students compared to secondary students, possibly because high-stakes exams reduce the enjoyment of gameplay (Cai et al. 2022). Similarly, Belland et al. (2017) observe comparable trends in STEM education, underscoring the need for further research to understand DGBL's effectiveness for secondary learners.

Beyond its influence on learning outcomes, scaffolding has also been examined through the lens of cognitive load theory, which describes three types of cognitive load competing for limited working memory. *Intrinsic* cognitive load arises from the inherent complexity of the learning content, *germane* cognitive load refers to cognitive resources directed towards deep processing and learning, and *extraneous* cognitive load involves information or activities irrelevant to the learning task (Paas et al. 2004). Effective instructional design aims to minimise extraneous cognitive load, allowing learners to focus cognitive resources on deep learning (Skulmowski and Xu 2022). In DGBL, learners acquire knowledge from navigating and interacting with game elements, which may place a high demand on their cognitive processes. For example, some game elements can slow learners down and distract them from learning, imposing a high extraneous load on their limited working memory and negatively impacting learning (Cai et al. 2022). Other elements may direct learners' attention towards relevant portions of the game or encourage learners to reflect on their game moves, thereby increasing intrinsic or germane load (O'Neil et al. 2014). While scaffolding in DGBL should aim to reduce extraneous load, some studies report that it has no significant effect on cognitive load (e.g., O'Neil et al. 2014; Schrader and Bastiaens 2012; Zumbach et al. 2020). Moreover, some researchers caution that scaffolds may increase extraneous load, disrupt game flow and hinder deep processing and learning (Cai et al. 2022). For example, Adams and Clark (2014) found that students who received self-explanation prompts in a science game performed worse than those who did not, suggesting that the prompts may have imposed a high cognitive burden that interfered with learning. These findings highlight the need for further research to explore the interplay between scaffolding, cognitive load and learning outcomes in DGBL.

Cai et al. (2022), in their meta-analysis of the effects of scaffolding on student achievement in DGBL, emphasise the importance of investigating adaptive scaffolding, which has become increasingly feasible with technological advancements. They also called for further analysis of learners' behavioural patterns during gameplay to better understand engagement patterns. In light of the literature, the present study investigated two types of scaffolds in a science game for secondary students: a soft, adaptive scaffold implemented through a GenAI-supported chatbot and a hard, expository scaffold via instructional videos. To assess the impact of these scaffolds, the study evaluated not only science learning outcomes but also cognitive load and student behaviours during gameplay. The following sections provide an overview of the literature on these two types of scaffolds in the context of DGBL.

2.2 | Instructional Videos as a Hard Scaffold in DGBL

Videos have increasingly become a key instructional modality in education today. They are often central in flipped classrooms (Akçayır and Akçayır 2018) and are a staple in MOOCs (Fidan and Gencel 2022). In addition to delivering instructional content, videos can also serve as tools for reflection or feedback (Sablic et al. 2021). Compared to instructional texts, videos convey information using a combination of visuals and audio, engaging both auditory and visual processing channels (Mayer 2009; Noetel et al. 2021). Video-based learning has been shown to enhance learner engagement, motivation and performance across a variety of settings (Noetel et al. 2021; Sablic et al. 2021).

Digital games are characterised by their reliance on video as a primary modality. In the context of DGBL, engaging videos can enhance storytelling, heighten relevance and interest and create a more immersive narrative experience (Ke 2016). Videos can also provide just-in-time instruction that supports learners at key moments in-game play (Ke 2016). There are two primary types of instructional videos used in DGBL. The first type demonstrates problem-solving processes within the game itself (Cai et al. 2022). For example, in a science puzzle game, when students struggle with a puzzle, they can access a video that demonstrates the solution steps (Kao et al. 2017). The second type of video, which we focus on in this study, is not situated directly with the game context but instead elucidates disciplinary knowledge relevant to the game. For example, if students encounter difficulties with a task involving speed and distance, they can watch a video explaining these concepts and their interrelations (Liao et al. 2019). In this study, we define these as concept-focused instructional videos that provide on-demand instructions, akin to an ever-present teacher who supports learners as they engage in game-based inquiries (Shin et al. 2020). Such videos can aid learners in bridging intuitive

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gameplay experiences with formal disciplinary knowledge (Liao et al. 2019).

Despite the promise of instructional videos as scaffolds in DGBL, there is limited empirical research examining their effects, particularly the impact of concept-focused videos. Liao et al. (2019) is the only study we identified in the literature, reporting that instructional videos explaining physics concepts within DGBL significantly improved learning outcomes but had no effect on cognitive load. Further research is needed to understand how these types of instructional videos may impact science learning outcomes, cognitive load and in-game behaviours.

2.3 | GenAI-Supported Chatbot as an Adaptive Soft Scaffold in DGBL

Researchers have long experimented with adaptive scaffolding to provide individualised learning support, spanning from intelligent tutoring systems (Anderson et al. 1985), on-demand scaffolds by human tutors (Azevedo et al. 2004), to peer scaffolding. Recent advancements in education data mining and artificial intelligence (AI) have opened new possibilities for adaptive scaffolds. For example, Lim et al. (2023) used a rule-based system to adaptively scaffold college students' self-regulated learning (SRL). Among the various types of AI-powered scaffolding tools, this study focuses on chatbots. By leveraging AI and natural-language processing, chatbots engage users in verbal or written conversations (Yin et al. 2021), enabling adaptive, soft scaffolding. Chatbots have been used across various disciplines to scaffold learning, with mixed findings regarding their effectiveness. While some studies report that chatbots improve learning outcomes and self-efficacy (e.g., Chang et al. 2021; D'Mello and Graesser 2013; Essel et al. 2022; Lee et al. 2022; Yetişensoy and Karaduman 2024), others have found no significant impact (e.g., Han et al. 2022; Topal et al. 2021; Yin et al. 2021). Although few studies have explored the impact of chatbots on cognitive load, Brachten et al. (2020) found that a chatbot-enabled virtual assistant reduced college students' cognitive load in a problem-solving task, and Abbasi et al. (2019) observed that students using a chatbot for programming assistance reported higher germane cognitive load compared to those relying on traditional search engines.

The existing literature highlights several themes regarding chatbots in education. First, chatbots are more extensively studied in some subjects, such as language learning (Wollny et al. 2021; Wu and Yu 2024), than in science education (Topal et al. 2021). Second, SRL may play an important role in learning with chatbots. A recent meta-analysis by Wu and Yu (2024) revealed that chatbots had a larger effect in higher education than in elementary and secondary education. Wu and Yu (2024) reasoned that lower SRL levels among younger learners may lead to less effective interactions with chatbots. A recent study by Chiu, Moorhouse, et al. (2023) provides some evidence supporting this conjecture. When comparing two groups of secondary ESL students using a chatbot for an English writing task, with or without additional teacher support, Chiu, Moorhouse, et al. (2023) found that students with lower SRL and no teacher support reported the lowest competence and motivation. Chiu, Moorhouse, et al. (2023) suggested that students with lower SRL might face particular challenges in interacting with chatbots, such as needing more accurate responses or being more likely to give up when experiencing confusion or failure. While writing performance was not measured, the findings suggest that SRL skills may be critical to the effectiveness of chatbot-supported learning. Further, in examining fifth-grade students' interaction logs with a chatbot to learn physics, Topal et al. (2021) found that students' questions were centred around key concepts but did not delve into subtopics. This suggests that students might lack deeper reflections on science concepts and the conceptual relationships among them. Overall, the studies highlight the importance of SRL for young learners as they interact with chatbots. Moreover, a recent systematic review suggested that chatbots in existing studies provide uneven support across SRL dimensions, placing greater emphasis on learning resources and strategies than on goal setting and reflection (Guan et al. 2025).

Research on chatbots in DGBL is still limited, despite studies on other adaptive scaffolds in gaming contexts (e.g., Faber et al. 2023; Hooshyar et al. 2021; Ronimus et al. 2014; Serge et al. 2013). The advent of GenAI presents unique opportunities to adaptively support learners' inquiry processes in science games. GenAI-powered chatbots, like ChatGPT, use natural-language processing to generate coherent and contextually relevant responses, enabling more natural and intuitive interactions (Mogavi et al. 2024). Despite ongoing concerns over academic integrity, misinformation and potential superficial learning, discussions persist on the educational benefits of GenAI (e.g., Chiu, Moorhouse, et al. 2023; Chiu, Xia, et al. 2023; Lo 2023; Memarian and Doleck 2023; Montenegro-Rueda et al. 2023). For instance, Kim and Adlof (2024) discuss ChatGPT's potential to offer interactive, just-in-time and personalised support in constructivist learning environments like inquiry learning, while Rejeb et al. (2024) emphasise the essential role of learner inquiries when using ChatGPT for learning.

Empirical research on GenAI in education is still emerging (Memarian and Doleck 2023). For example, Pardos and Bhandari (2023) found that human-generated hints were more effective than ChatGPT-generated hints. Li and Xing (2021) trained ChatGPT to support MOOC learners in online forums, finding that it offered informational, emotional and community support comparable to humans. Similarly, Li (2023) applied ChatGPT in a flipped classroom, observing significant improvements in student performance, attitude and motivation. Yilmaz and Yilmaz (2023) reported that students learning programming with ChatGPT showed higher computational thinking skills, programming self-efficacy and motivation than those without it. In DGBL, anecdotal evidence suggests that ChatGPT can enhance the enjoyment of learning in math games (Mogavi et al. 2024), though experimental studies in this area remain scarce (Kim and Adlof 2024; Wu and Yu 2024). Thus, questions remain: Does a GenAI-supported chatbot improve science learning outcomes in DGBL? How does it affect cognitive load? Furthermore, could GenAI-supported adaptive scaffolds be even more effective when combined with hard scaffolds, like instructional videos? Topal et al. (2021) found that fifth-grade students preferred having both a chatbot and instructional videos in science learning. Fidan and Gencel (2022) observed that preservice teachers with both tools outperformed those with only instructional videos, though Han et al. (2022) found no

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additional benefit from using both. This study seeks to explore how a GenAI-supported chatbot and instructional videos may complement each other in DGBL. Specifically, we integrated a GenAI-supported chatbot as a scaffold within the gameplay. While chatbots can serve various purposes, such as promoting emotional well-being, this chatbot was specifically designed to enhance cognitive engagement through real time, natural language and personalised interactions. Its primary goal was to deepen students' conceptual understanding of the physics content embedded in the game.

2.4 | Synthesis of the Literature and the Current Study

DGBL has shown promise in enhancing learning and motivation (Lamb et al. 2018; Ryan and Rigby 2020; Vlachopoulous and Makri 2017), though its effectiveness varies across subjects and learner groups, with mixed results in science education (Girard et al. 2013; National Research Council 2011; Young et al. 2012). Scaffolding plays a crucial role in supporting student inquiry and knowledge acquisition in science games (Barzilai and Blau 2014; Wouters et al. 2013; Wouters and van Oostendorp 2013). However, the effects of different types of scaffolds, particularly on cognitive load, remain unclear (Adams and Clark 2014; Cai et al. 2022). This study focuses on two scaffold types: instructional videos as hard scaffolds and a GenAI-supported chatbot as a soft scaffold. Instructional videos provide structured, on-demand support by clarifying scientific concepts during gameplay (Liao et al. 2019; Ke 2016), while adaptive GenAI-powered chatbots offer personalised assistance, potentially enhancing inquiry and engagement (Kim and Adlof 2024; Yilmaz and Yilmaz 2023). Despite their potential, limited research has explored the individual and combined effects of these scaffolds in DGBL, particularly for secondary students who often face motivational challenges (Cai et al. 2022). To address these gaps, we implemented both scaffolds in a science game for secondary students and examined their effects on science knowledge acquisition, cognitive load, in-game performance and game behaviours. Through this study, we aim to provide educators with best practices for effectively integrating instructional videos and GenAI-supported chatbots into the DGBL curriculum for science education.

3 | Methodology

3.1 | Research Design and Participants

This study employs a 2×2 factorial experimental design to assess the impact of two scaffolds in a DGBL environment: instructional videos (presence or absence) and a GenAI-supported chatbot (presence or absence). The factorial design was appropriate for answering the research question, as it facilitated the investigation of both the individual and combined effects of instructional videos and the GenAI chatbot in DGBL. The experimental design is illustrated in Figure 1.

The participants in this study consisted of 160 seventh-grade students (aged 12–13; 73 females and 87 males) from the same

框架：2×2 實驗設計與流程

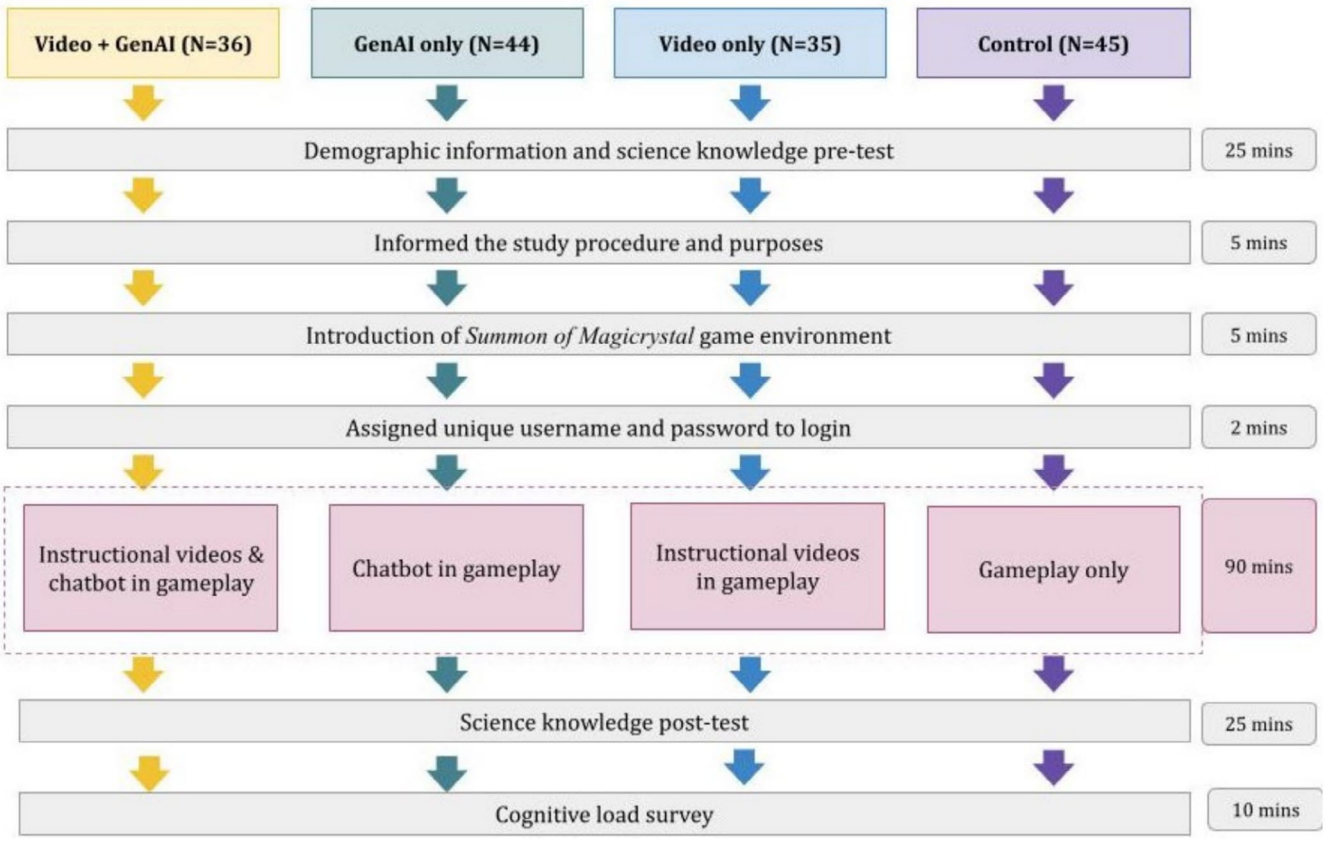


FIGURE 1 | Experimental design and procedure.

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school in central Taiwan. This demographic group was selected because seventh-grade is when students begin learning physics as part of their general sciences curriculum. Additionally, the learning content covered in DGBL aligned with the seventh-grade curriculum, ensuring its relevance to their educational level. A self-report survey indicated that the students had varying levels of prior gameplay experience: 25% classified themselves as beginner players, 40% as intermediate players, 30% as advanced players and 5% as professional players. Additionally, 75% of the participants reported having previously used online games for learning purposes. The content was designed to match their cognitive development stage, with the DGBL context adding elements of fun and engagement.

As illustrated in Figure 1, the participants were randomly assigned by intact classes into one of four experimental groups: video only (35 students), GenAI chatbot only (44 students), video + GenAI chatbot (36 students) and a control group without video or GenAI chatbot (45 students). Random assignment at the class level helped control potential differences between students while maintaining the integrity of the classroom settings.

3.2 | Materials

3.2.1 | DGBL Environment

Summon of Magicrystal is a computer-based adventure and simulation game designed for secondary students. The game immerses players in a village under attack by a dragon, where their objective is to rescue local villagers by securing vital life supplies in a timely manner while collecting magic crystals in the process. Newton's Laws of Motion are embedded in various game adventures, with each adventure's objectives aligned with the secondary science curriculum standards in Taiwan. Throughout the rescue missions, students engage in simulation activities, interacting with different elements in the game world where physics concepts such as time, distance and velocity are operationalised. For instance, as depicted in Figure 2, students manipulate a speed control (set here to '5') and use a timer to observe the movement of food items along a conveyor belt. The goal is to determine how long it takes each item to travel a given distance. As items move from left to right at the set speed, students use the stopwatch (displayed in the top right) to measure the time interval between the appearance of each item. They then refer to the ruler beneath the conveyor belt to measure the distance travelled. By adjusting the speed setting and observing the resulting changes in time and distance, students apply the physics formula $distance = speed \times time$ to infer actual speeds and make predictions about future motion. The game score, displayed in the top-right corner of the interface, offers immediate feedback on students' performance and progress. Each adventure features both basic and advanced levels, each presenting distinct challenges. Upon accumulating a certain number of points or completing specific tasks at the basic level, students can progress to the next level, unlocking more demanding challenges. They have the opportunity to attempt each level multiple times in order to advance in the game. All interactions with game elements are tracked and recorded by the game database.

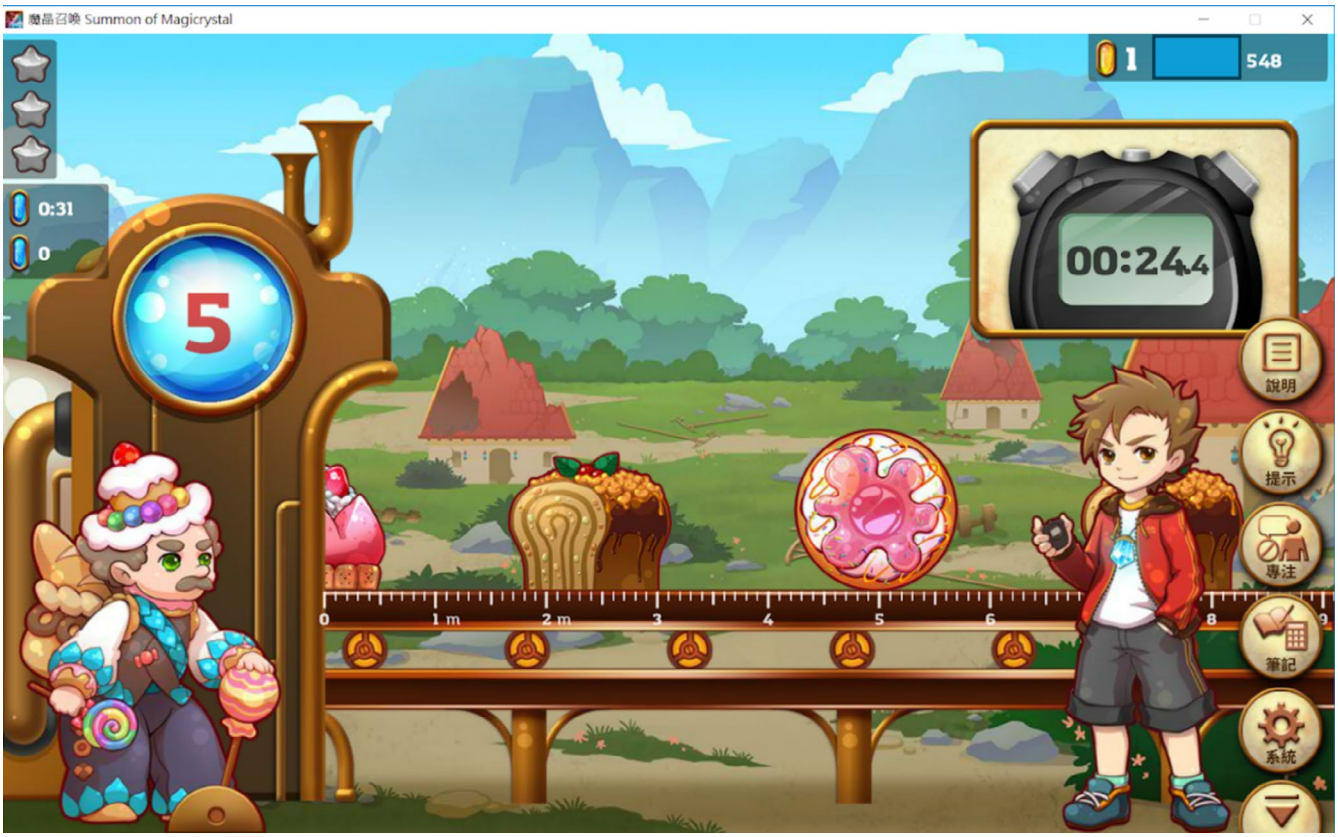


FIGURE 2 | The game interface, with features including (1) variable control through a speed metre and a timer to move food supplies; (2) game score; (3) notebook and calculator.

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FIGURE 3 | The game board.

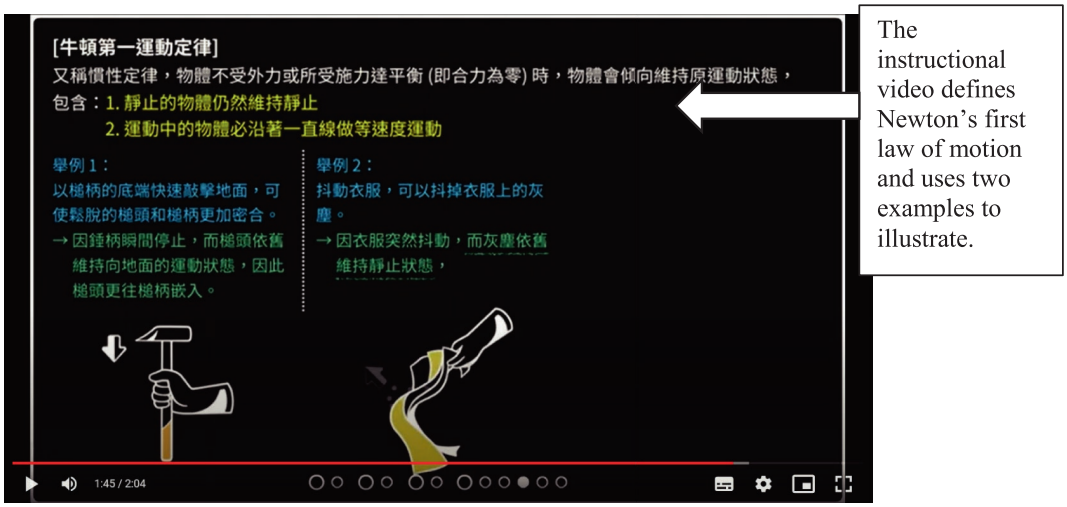


FIGURE 4 | An example instructional video in the game.

Throughout gameplay, students have access to several support features. A game board (as shown in Figure 3) can be accessed at any point to obtain information about the goals for the current level, procedural guidance and definitions of key science concepts. Additionally, a notebook and calculator tool (Figure 1) are available to students at any point during the game, allowing them to take notes or perform calculations. These features are available to all students, regardless of their treatment groups.

3.2.2 | Instructional Videos

For the two groups that received video scaffolds (video only and video + GenAI), an instructional video was provided for each adventure. These videos, created by two experienced science teachers, aimed to elucidate fundamental concepts of motion and force within each adventure. They incorporated a combination of textual descriptions, visual examples featuring graphs and animations and synchronised voice narration (Figure 4). The videos ranged from 1 to 3.5 min in duration. Students could access the videos upon entering an adventure or at any time during gameplay via the game board. They had the flexibility to pause a video or jump to specific segments at will.

3.2.3 | GenAI-Supported Chatbot

For the two groups that received GenAI scaffolds (GenAI only and video + GenAI), a chatbot companion was available during each adventure. The chatbot was powered by the ChatGPT version 3.5-turbo model. As shown in Figure 5, the architecture of the game system comprised three main components. At the backend, the GPT-3.5-turbo model provided a chatbot conversational AI service to generate text messages, enabling students to engage in dynamic text-based interactions at the frontend. The

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middle layer acted as an intermediary between the frontend and backend, facilitating user-system interaction. The chat history recorded users' questions and the messages exchanged during the interaction.

Students could engage in real-time conversations with the chatbot using natural language at any point while navigating the tasks and challenges in the game. As illustrated in Figure 6, students could choose from a list of recommended questions. These included prompts such as 'When does inertia occur in the game?' and 'How can I achieve a high score?' Alternatively, students could also ask their own questions. The chatbot was not designed to provide direct answers but to engage students in problem-solving and critical thinking through tailored guidance and hints contextualised to their specific gameplay situations. For example, if a student asked, 'Why is my solution not working?' the chatbot might analyse their input and respond, 'Check whether you have accounted for all variables in the equation. Remember, the value of X changes with each iteration'. Contextualised responses were generated during runtime through system prompting. Specifically, the system dynamically generated prompts based on level-specific content and students' real-time gameplay progress. These prompts guided the chatbot's responses, which were further tailored by incorporating students' input or questions. An excerpt is provided in Appendix B to demonstrate how the chatbot interacts with students based on their specific gameplay context.

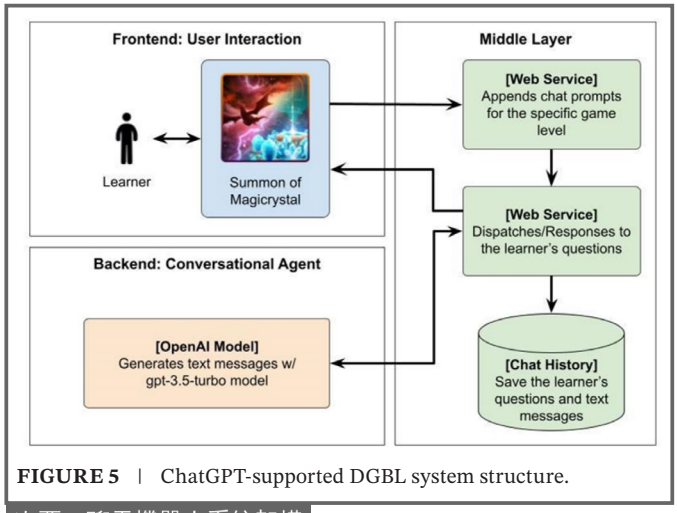


FIGURE 5 | ChatGPT-supported DGBL system structure.

次要：聊天機器人系統架構

3.3 | Data Sources

3.3.1 | Science Knowledge Test

Students' understanding of the motion and force concepts addressed in the game was assessed using a test adopted from Liao et al. (2019). The original test was developed by two experienced science teachers and validated by a university professor (Liao et al. 2019), with Cronbach's alpha reported as 0.875. For the current study, two experienced science teachers reviewed the relevance of the test to the game content to ensure its content validity. The test consists of 15 items, including 13 multiple-choice questions and two fill-in-the-blank questions. Cronbach's alpha for the test was 0.81, indicating good reliability (Nunnally and Bernstein 1994). Test scores were calculated based on the percentage of correctly answered questions, with a total possible score of 100 points. Sample questions can be found in Appendix A.

3.3.2 | Cognitive Load

Students' cognitive load was measured using an instrument adapted from Leppink et al. (2013). Previous studies employing the instrument report good to excellent reliability, along with evidence supporting its construct and predictive validity (e.g., Leppink et al. 2013; Zukić et al. 2016). For the current study, the original instrument was translated into Chinese, and the wording of certain items were adjusted to fit the DGBL context. For instance, references to 'activity' in the original questionnaire were replaced with 'game'. The adapted instrument included 10 items: three measured intrinsic cognitive load (e.g., 'The topics covered in this game were very complex'), three measured extrinsic load ('The instructions during the gameplay were very unclear') and four measured germane



FIGURE 6 | GenAI-supported chatbot in the game.

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load ('The gameplay really enhanced my understanding of the topic(s) covered'). Students rated their agreement with each statement using a 5-point Likert scale, ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). The reliability of the three subscales, as measured by Cronbach's alpha, was 0.91, 0.89 and 0.81 respectively, indicating good to excellent reliability (Nunnally and Bernstein 1994).

3.3.3 | In-Game Performance

Students' in-game performance was assessed using two metrics. The first metric is the total points accumulated from gameplay by completing specific tasks or reaching certain milestones. These total points serve as a quantitative measure of students' success in achieving game objectives. The second metric is the total number of attempts made throughout the game duration. Each attempt represents a new opportunity for students to refine their strategy, learn from past mistakes and advance further in the game.

3.3.4 | Game Behaviours

Throughout their gameplay, each move made by students was tracked and recorded by the game database. For instance, students could access the game board to review information or use a specific game object to manipulate a variable (e.g., adjusting speed using speed control). Consequently, each student left behind a sequential trace of game behaviours, to which we assigned different codes (see Table 1). For example, the behaviour of visiting the game board was assigned the code 'BR'. As a result, students' game behaviours were converted into a string of codes, such as BR-VO-LU. Since the codes were based on objective game behaviours (e.g., accessing the game board), they provide a reliable measure, minimising potential biases from subjective interpretation. Nevertheless, we implemented measures to ensure data integrity by verifying that all logged actions and behavioural sequences were accurately captured, with no missing or corrupted entries. Similar behavioural codes and processes have also been applied and validated in Chen and Chang (2024).

TABLE 1 | Codes for game behaviours.

Code	Behavioural category	Descriptions
BR	Board reading	Students visit the game board to access game level objectives, procedural guidance and definition of key concepts.
VO	Variable operation	Students interact with game elements to manipulate variables for desired effects.
LU	Level upgrade	Students complete a game level or proceed to the next level.
NU	Note use	Students use tools (a notebook or a calculator) to record data/ideas or carry out calculations.
AI	GenAI chatbot	Students activate GenAI chatbot to get immediate answers to questions, explanations of complex concepts or step-by-step guidance for solving problems.
IV	Instructional video	Students access instructional videos to gain a systematic understanding of relevant science concepts.

3.4 | Procedure

The study received institutional review board (IRB) approval prior to its initiation. Following institutional regulations and ethical guidelines, informed consent was obtained from both students and their guardians, who were informed about the study's purpose, procedures and participants' right to withdraw at any time. To ensure confidentiality, participants were assigned unique, random usernames and all data, including students' chat logs with the chatbot, were anonymised to prevent personal identifiers from being linked to the data, including the chat logs. All data were securely stored on password-protected systems with access restricted to authorised personnel. Additionally, data security measures, such as encryption and secure storage, were implemented to protect participant information throughout the study.

The study spanned 3 weeks. At the beginning of Week 1, participants completed a science knowledge pretest and a demographic survey. The intervention began in Week 2. Researchers first briefed students on the study's purpose and procedures, introduced the game environment, and provided each participant with a unique username and password. Students in the two GenAI groups received additional instructions on how to use the GenAI chatbot within the game. The intervention comprised two 45-min sessions, totalling 90 min. During these sessions, students engaged in gameplay according to their assigned treatment conditions. To isolate the treatment effects, the game environment, elements, features and available actions were kept identical across all four groups. The only variation was the configuration of video and chatbot scaffolds, based on the specific conditions described earlier. In Week 3, participants took the science knowledge posttest (identical to the pretest) and completed the cognitive load survey. Figure 1 outlines the entire experimental procedure, including the duration for each phase.

3.5 | Data Analysis

The Statistical Package for the Social Sciences (SPSS 24 for Windows) was used for all statistical analysis. Descriptive statistics and correlations were first computed for the relevant variables. To assess the treatment effects on students' science knowledge, we conducted a 2×2 factorial ANCOVA, using the science knowledge pretest score as a covariate. Compared to calculating pre-post gain scores on the knowledge test, this approach allowed us to examine the main effects of the two

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scaffolds, as well as their interaction effects on students' science knowledge, while controlling for any pre-existing differences in knowledge prior to the intervention. This resulted in greater statistical power, reduced error variance and more precise estimates of treatment effects (Dimitrov and Rumrill 2003; Senn 2006).

For the treatment effects on cognitive load and in-game performance, we conducted separate 2×2 factorial ANOVAs for each dependent variable. Prior to running these analyses, we conducted preliminary tests to ensure that the assumptions of the statistical models were met. Box's M Test was used to check the equality of covariance matrices and Levene's Test was performed to assess the homogeneity of variance. The results indicated that both assumptions were satisfied at the 0.05 significance level, confirming the appropriateness of using ANCOVA and ANOVA for the data analysis.

To analyse students' game behaviours, we conducted lag sequential analysis using General Sequential Querier (GSEQ) software (Bakeman and Gottman 1997). This method enabled us to examine the coded sequential behaviours for each group, identifying significant behavioural patterns and transitions over time. The lag sequential analysis provided deeper insights into the interaction dynamics within the DGBL environment.

4 | Results

4.1 | Descriptive Statistics and Correlations

Table 2 presents the descriptive statistics for the science knowledge pretest and posttest, cognitive load and in-game performance. The science knowledge posttest ranged between 44.03 and 63.33 across the four groups, comparable to the 44.83–66.60 range reported in previous research using a similar assessment (Liao et al. 2019). The ranges of the three cognitive load measures are largely comparable with those reported in previous research using the same instrument, though under different treatment conditions (Chen and Chang 2024; Liao et al. 2019). For intrinsic load, scores ranged from 3.31 to 3.78 across the four treatment groups, closely aligning with the 3.24–3.74 range reported by Liao et al. (2019) and Chen and Chang (2024). Germane load scores ranged from 2.85 to 3.38, within the comparative range of 2.37–3.29 found in prior studies. Extraneous load ranged from 2.41 to 3.03, with the comparative range in previous studies being 1.18–3.02; the lower bound of 1.18 reflects the differing treatments in those studies. The correlations among the variables are shown in Table 3.

4.2 | Treatment Effects on Science Knowledge

An ANOVA comparing the pretest scores of the four groups revealed no significant difference, $F = 2.10$, $p = 0.11$. A 2×2 factorial ANCOVA on the science knowledge posttest, with pretest scores as a covariate, showed a significant main effect of instructional videos, F (1, 153) = 55.64, $p < 0.001$, partial $\eta^2 = 0.27$, observed power = 1.00, indicating a large effect size (Cohen 1988). Both groups with video scaffolds (video + GenAI and video only) significantly outperformed the two groups without instructional videos (GenAI only and Control). There was no main effect of GenAI, nor was there an interaction effect, suggesting that the combination of instructional videos and GenAI chatbot did not improve students' performance more than either video or chatbot alone.

4.3 | Treatment Effects on Cognitive Load

A 2×2 factorial ANOVA was conducted on each of the cognitive load subscales. Results indicated a significant Video×GenAI interaction effect on extraneous cognitive load, F (1, 153) = 6.75, $p = 0.01$, partial $\eta^2 = 0.04$, observed power = 0.73. No other significant effects were observed. As depicted in Figure 7, GenAI scaffolds had a differential effect on extraneous load depending on the presence of instructional videos. Specifically, when videos were absent, GenAI increased extraneous load, as illustrated by the dotted line in the figure. However, when instructional videos were present, GenAI scaffolds significantly reduced the load, as indicated by the solid line. The combination of both GenAI and instructional videos resulted in the lowest extraneous load.

結論：知識、負荷、表現描述統計

TABLE 2 | Descriptive statistics.

Variable	Video + GenAI ($N = 36$)	Video only ($N = 35$)	GenAI only ($N = 44$)	Control ($N = 45$)
Science knowledge test				
Pretest	34.25 (16.85)	40.38 (12.09)	39.84 (16.42)	41.67 (9.70)
Posttest	63.33 (18.75)	63.24 (17.95)	48.18 (14.15)	44.03 (11.99)
Cognitive load				
Intrinsic	3.31 (1.07)	3.57 (0.96)	3.46 (1.06)	3.78 (0.94)
Extraneous	2.41 (0.91)	3.01 (0.95)	3.03 (1.01)	2.83 (0.82)
Germane	3.38 (1.07)	3.15 (1.09)	2.85 (1.08)	3.15 (1.00)
In-game performance				
Game attempts	4.89 (2.92)	6.34 (7.00)	2.89 (3.00)	7.98 (6.00)
Game points	498.66 (304.92)	498.26 (358.45)	415.18 (320.06)	482.26 (540.15)

Note: $N = 160$.

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TABLE 3 | Correlations among the study variables.

	1	2	3	4	5	6
1. Science knowledge posttest	1	−0.06	−0.23**	0.15	−0.11	0.24**
2. Intrinsic cognitive load		1	0.38**	−0.04	0.12	−0.03
3. Extrinsic cognitive load			1	−0.46**	0.11	−0.01
4. Germane cognitive load				1	0.08	0.12
5. Game attempts					1	0.56**
6. Game points						1

Note: $N = 160$.
**$p < 0.01$.

結論：影片×GenAI 對外在認知負荷

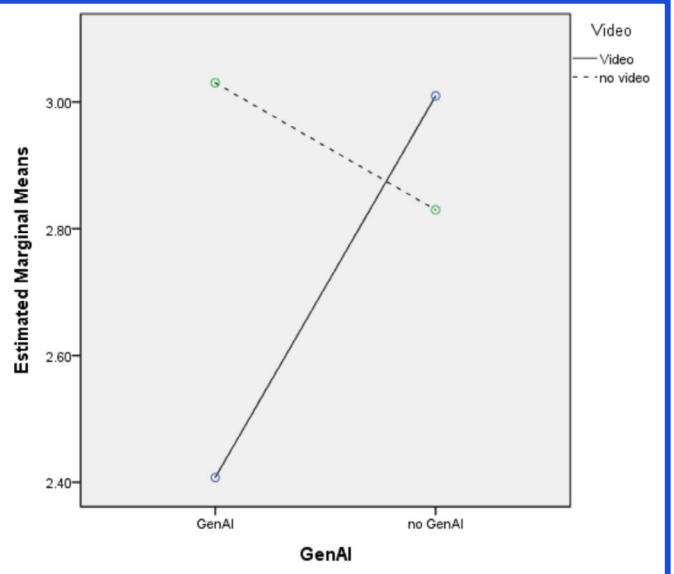


FIGURE 7 | Video×GenAI interaction effect on extraneous cognitive load.

次要：遊戲嘗試次數的交互作用

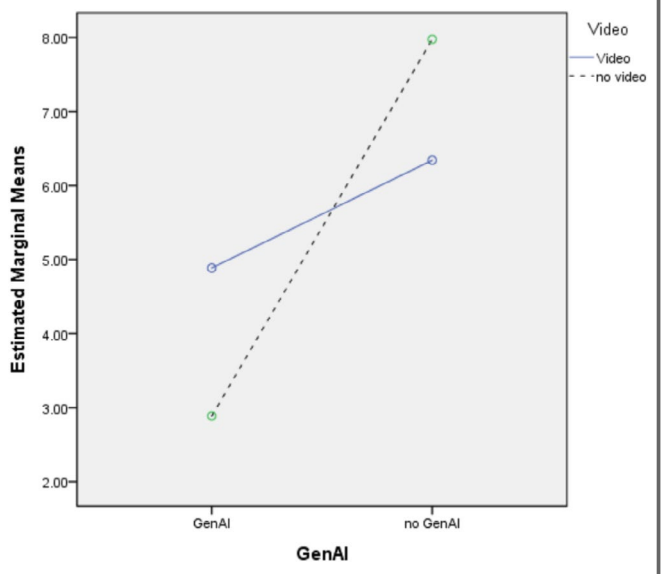


FIGURE 8 | Video×GenAI interaction effect on game attempts.

4.4 | Treatment Effects on In-Game Performance

We conducted 2×2 factorial ANOVAs on the two metrics of in-game performance. The results revealed a significant main effect of GenAI on game attempts, $F(1, 153) = 16.53$, $p = 0.001$, partial $\eta^2 = 0.09$, observed power = 0.98, indicating a medium effect size (Cohen 1988). The groups with GenAI scaffolds made significantly fewer attempts than the other two groups. Additionally, a significant interaction between video and GenAI scaffolds was found for game attempts, $F(1, 153) = 5.09$, $p = 0.03$, partial $\eta^2 = 0.03$, observed power = 0.61. As depicted in Figure 8, the effect of GenAI on game attempts was markedly stronger when instructional videos were absent. In these conditions, participants using GenAI scaffolds significantly reduced the number of attempts made during the game, as shown by the dotted line in the figure. In contrast, the impact of GenAI scaffolds on game attempts was less pronounced among the two groups with access to instructional videos, as indicated by the corresponding dotted line.

4.5 | Treatment Effects on Game Behaviours

Descriptive statistics of behavioural frequencies across the four treatment groups are presented in Table 4. Lag sequential analysis revealed different behavioural patterns among the four groups. Figure 9 illustrates the significant behavioural sequences of each group, represented by nodes (e.g., VO) and arrows linking them. In Figure 9a, the control group shows three primary patterns. First, students repeatedly engaged in the four game activities—reviewing the game board (BR), adjusting variables (VO), using the notebook/calculator tool (NU) and levelling up (LU)—as denoted by the circular arrows on each node. Second, the LU → BR → VO → LU sequence shows that after levering up, students reviewed the game board to familiarise themselves with the level objectives, game procedures and key science concepts. They then engaged in simulations by adjusting variables, observing the effects and making modifications to achieve desired outcomes. After they had accumulated sufficient points, they advanced to the next level and repeated the sequence. The third pattern, a two-way BR-NU connection, indicates that students frequently alternated between reviewing the game board and using the notebook/calculator tool.

The video only group (Figure 9b) exhibited the same three patterns as the control group but also showed three additional ones. First, as indicated by the circular arrow on the IV node, students repeatedly watched instructional videos. Second, the two-way IV-BR connection suggests repeated alternations

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between watching videos and reviewing the game board. Since students accessed instructional videos through the game board, this sequence was expected. However, the frequent switches might have prompted students to intentionally connect the instructional videos with the information on the game board (level objectives, game procedures and key concepts). The third new pattern was the BR-LU sequence, which was unique to this group. While all other groups only showed the reverse sequence (LU → BR), the video only group displayed a two-way BR-LU connection, which is less intuitive.

TABLE 4 | Means and standard deviations of behavioural frequencies by treatment group.

	Control	Video only	GenAI only	Video + GenAI
BR	26.24 (18.72)	13.37 (10.28)	7.48 (4.84)	9.61 (4.76)
VO	454.80 (393.86)	331.19 (168.98)	236.15 (129.38)	196.79 (110.30)
NU	8.74 (12.72)	10.34 (9.87)	5.02 (5.12)	8.53 (7.38)
LU	2.43 (3.90)	2.43 (3.90)	2.06 (1.35)	2.03 (1.36)
IV	—	2.37 (1.15)	—	2.83 (1.44)
AI	—	—	4.15 (3.17)	4.92 (4.44)

Note: $N = 160$.
Abbreviations: AI, GenAI chatbot; BR, board reading; IV, instructional video; LU, level upgrade; NU, note/calculator use; VO, variable operation.

Typically, students review the game board after levelling up, but doing so before completing or starting a new level was unusual. More nuanced coding is required to understand the underlying reason for this behaviour.

The GenAI only group (Figure 9c) shared the same first two patterns as the control group: (1) repeated engagement in BR, VO, NU and LU activities, and (2) the LU → BR → VO → LU sequence. However, with the GenAI chatbot available, the BR-NU connection disappeared, indicating that students significantly reduced the alteration between reviewing the game board and using the notebook/calculator tool. Instead, they frequently interacted with the chatbot, as indicated by the circular arrow on the AI node. However, both AI and NU nodes became disconnected from other game behaviours, suggesting that students used the chatbot and notebook/calculator in isolation, without integrating them into a coherent sequence. Overall, this group exhibited the most fragmented behaviour patterns among the four groups.

Lastly, the video + GenAI group (Figure 9d) displayed distinct behavioural patterns compared to the other groups. Unlike the video only group, there was no longer a BR-NU connection, indicating that students did not alternate between reviewing the game board and using the notebook/calculator. Instead, the GenAI chatbot acted as an intermediary between the two. Specifically, students frequently switched between the chatbot and the notebook/calculator (AI-NU), likely in response to conversations with the chatbot. After interacting with the chatbot, students often revisited the game board, where they could also access instructional videos (AI → BR → IV), suggesting that students were likely to reorient themselves to the level objectives and game tasks on the game board, and may subsequently watch

結論：四組遊戲行為序列

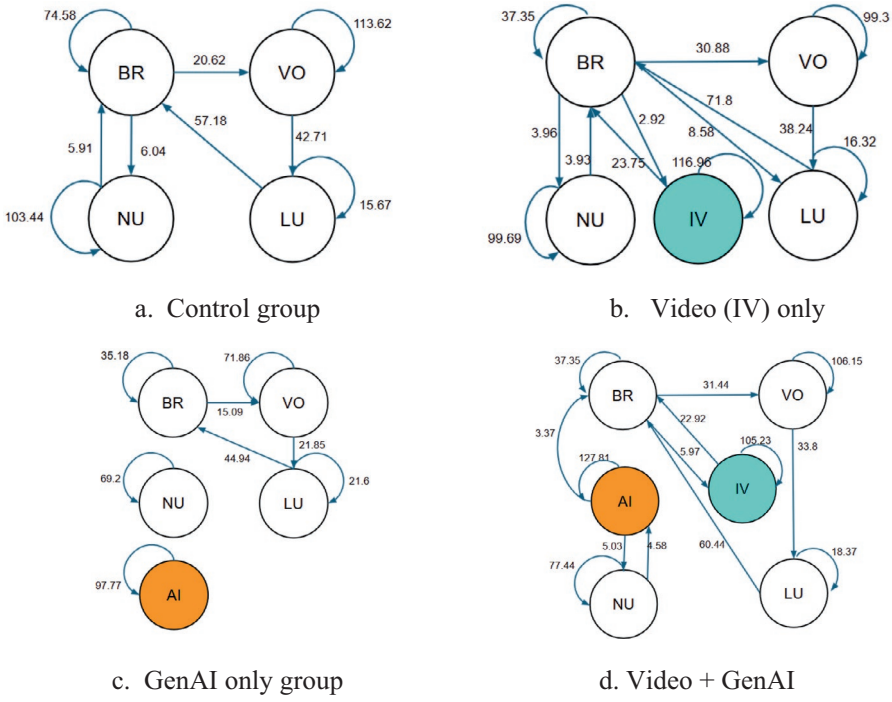


FIGURE 9 | Sequential patterns of the four treatment groups (AI, GenAI chatbot; BR, board reading; IV, instructional video; LU, level upgrade; NU, note/calculator use; VO, variable operation).

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instructional videos. In comparison to the GenAI only group, the AI node in this group was no longer isolated but closely integrated with both BR and NU, suggesting that the chatbot was more embedded in the gameplay. Overall, this group exhibited more connected and cohesive behaviours than the fragmented patterns observed in the GenAI only group.

5 | Discussion

In a DGBL environment for secondary science, this study explored the impact of two distinct scaffolding strategies on science learning outcomes, cognitive load, in-game performance and game behaviours. The findings provide valuable insights into the effectiveness of instructional videos and GenAI-supported chatbots to enhance science learning within a DGBL context, which are discussed in the sections below.

5.1 | Instructional Videos in DGBL

Video-based learning has been recognised for enhancing engagement, motivation and learning outcomes (Noetel et al. 2021; Sablic et al. 2021). However, despite being a defining feature in video games, the role of instructional videos as a learning scaffold within DGBL environments has received limited examination. In this study, two groups of students were given access to instructional videos that elucidated the science concepts embedded in the game. The findings revealed a significant main effect of the instructional videos on students' science learning outcomes. Given the large effect size, these videos served as a strong scaffold, enhancing students' comprehension of the targeted science concepts. On the other hand, the videos did not exert a significant impact on cognitive load and game performance. This outcome aligns with the work of Liao et al. (2019), who also reported a positive influence of instructional videos on science learning in DGBL but found no significant effect on cognitive load. In accordance with recent systematic reviews on the efficacy of video-based learning across various learning contexts (e.g., Noetel et al. 2021; Sablic et al. 2021), our results offer empirical evidence for the utility of videos as a scaffolding tool in science games. Thus, instructional videos are beneficial not only for guiding students through game-specific tasks, like solving puzzles (Kao et al. 2017), but also for conveying disciplinary concepts embedded within a game without imposing additional extraneous cognitive load on learners.

Drawing from our findings and the existing literature, we propose three potential mechanisms that might explain the effectiveness of concept-oriented instructional videos in DGBL. First, the video format may enhance cognitive processing more effectively than instructional texts. In our study, although the game board provided descriptions of the same science concepts to all groups, the two groups with access to videos demonstrated superior performance. This suggests that the dual channels of auditory and visual information present in the instructional videos could clarify complex scientific concepts more effectively than text alone (Mayer 2009; Noetel et al. 2021). Second, the video format aligns more closely with the preferences of today's learners. Students today are increasingly exposed to video content, not only in educational settings but also through social media and other daily interactions (Fidan and Gencel 2022). This widespread exposure may make students more receptive to and engaged with learning through instructional videos. Third, unlike traditional expository science learning, which typically adheres to a 'learn first, apply next' approach, on-demand instructional videos provide timely support during the inquiry process. Our sequential analysis showed that students frequently revisited the videos and alternated between watching them and visiting the game board. Although the videos were not contextualised in the game narrative and context, they provided structured and systematic explanations of science concepts. As students alternated between viewing the videos and visiting the game board, they likely drew connections between the abstract concepts presented in the videos and game elements and mechanics displayed on the game board, leading to improved performance.

5.2 | GenAI-Supported Chatbot in DGBL

Scaffolding, by its very definition, needs to be tailored to meet the individual needs of learners (Wood et al. 1976). For decades, researchers have explored various approaches to adaptive scaffolding (e.g., Anderson et al. 1985; Azevedo et al. 2004). In more recent developments, AI-powered chatbots have shown promising capabilities (e.g., Lim et al. 2023). The advent of GenAI technologies, such as ChatGPT, offers even greater potential for adaptive scaffolding by facilitating interactions between learners and chatbots in a manner that is both coherent and contextually relevant (Mogavi et al. 2024). Despite the growing discourse on the potential benefits of GenAI in education, empirical studies, particularly in the DGBL context, remain in their infancy. In this research, we introduced a GenAI-supported chatbot as a scaffold within a DGBL environment. This allowed students to seek contextualised tips, suggestions and feedback while engaging in science inquiries with the game.

The outcome from our experimental study did not indicate a significant effect of the chatbot on science learning outcomes, cognitive load or game performance. This contrasts with the findings of Li (2023) and Yilmaz and Yilmaz (2023), who reported a positive influence of ChatGPT on learning outcomes. When compared with research on earlier types of chatbots, our results diverge from several studies that identified benefits of chatbots for learning (e.g., Chang et al. 2021; D'Mello and Graesser 2013; Essel et al. 2022; Lee et al. 2022), but align with others that did not observe a significant impact (Han et al. 2022; Topal et al. 2021; Yin et al. 2021). Regarding the chatbot's impact on cognitive load, our findings differ from those of Abbasi et al. (2019) and Brachten et al. (2020), who reported significant effects of chatbots on cognitive load in non-DGBL environments. The absence of a main effect of the GenAI chatbot on cognitive load in our study suggests that interactions with GenAI generally did not impose substantial cognitive demands on students. This contrasts with Clark et al. (2014) who suggest that self-explanation prompts might significantly increase cognitive load, hindering student performance in DGBL. Thus, compared with other forms of scaffolding, GenAI chatbots may offer scaffolding without overloading students' cognitive capacities during science inquiries in DGBL settings.

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Connecting with the existing literature, we propose three conjectures based on our findings. First, the absence of a significant impact from the GenAI chatbot might stem from the challenges seventh-grade students faced in using the chatbot effectively, particularly to support inquiry learning within a game-based environment. Existing studies that demonstrate positive effects of chatbots often feature college students as participants (e.g., Chang et al. 2021; Essel et al. 2022; Lee et al. 2022; Li 2023; Yilmaz and Yilmaz 2023), while research on K–12 learners, albeit limited, tends to show a lack of impact (e.g., Topal et al. 2021). Wu and Yu (2024) posited a similar conjecture in their meta-analysis, which found that chatbots had a larger effect in higher education compared to K–12 settings. They suggested that the disparity could be due to K–12 learners' insufficient SRL skills. In our study, for students to learn from gameplay, they had to engage strategically with simulations, navigate purposefully between the tangible game environment and abstract science concepts, and thoughtfully extract conceptual understandings from their experiences (Barzilai and Blau 2014; Wouters and van Oostendorp 2013). This inquiry process demands SRL skills, particularly the metacognitive processes of planning, monitoring and evaluating knowledge and strategies, which have been shown to influence science learning outcomes (Sinatra and Taasoobshirazi 2011). Although the GenAI chatbot in this study was capable of providing contextually relevant assistance, this support depended upon the students' inquiries. As researchers have underscored, the essence of learning with ChatGPT lies in the inquiry process, which involves formulating and posing prompts, thinking through these prompts and conversations with the chatbot, and challenging and clarifying information in the process (Rejeb et al. 2024; Rospigliosi 2023). Although the chatbot in our study was designed to guide students rather than provide direct solutions, it may not have offered sufficient SRL support to guide the inquiry process. When students lack SRL skills, their prompts may lack intentionality and strategy, similar to those observed in Topal et al. (2021), where students focused on main concepts but failed to engage in deeper inquiries about the concepts and their relationships. As a result, chatbot responses to student inquiries may be fragmented and insufficient for facilitating systematic and sustained inquiry. When these interactions fail to help students successfully complete game tasks, they may become frustrated and even give up (Chiu, Moorhouse, et al. 2023). This, in turn, impedes students' progressive construction of mental schemas regarding science concepts and their causal relationships. Although no study has directly investigated the influence of SRL in learning with GenAI-supported chatbots, research by Yilmaz and Yilmaz (2023) offers some insights. They found that students who utilised ChatGPT effectively for learning programming adopted a more strategic approach in employing the tool. Additional SRL support, such as a guided inquiry template embedded in the chatbot, could enhance its effectiveness, particularly within constructivist learning environments such as DGBL. Future research should test this hypothesis by incorporating direct SRL measures and examining their relationships with chatbot use and learning outcomes.

Our second conjecture concerns different uses of chatbots in learning. Some studies examine chatbots as substitutes for teaching (Chang et al. 2021; Essel et al. 2022; Lee et al. 2022; Yin et al. 2021), while others use them supplementary to teaching (Han et al. 2022; Li 2023; Topal et al. 2021; Yilmaz and Yilmaz 2023). When comparing the performance of those who learned from didactic sources with those who had additional chatbot support, existing research often suggests a lack of difference (Han et al. 2022; Topal et al. 2021). Similarly, we found no significant effect when the chatbot was supplementary to didactic instructional videos. Interestingly, among the few studies that found a positive impact of chatbots on K–12 students, the chatbot in D'Mello and Graesser (2013) offered a combination of didactic instruction and adaptive support. The relative impact of didactic instruction versus GenAI chatbots' adaptive scaffolding warrants further investigation.

Our third conjecture considers the nascent stage of GenAI applications in education, which might have triggered a novelty effect in our study. Huang et al. (2022) raised similar considerations in their systematic review of chatbot-supported language learning. In our study, the two GenAI groups made significantly fewer game attempts than the two groups without the chatbot. This may be due to a novelty effect, leading students to focus excessively on interacting with the chatbot at the expense of deeper engagement with learning and inquiry. The GenAI only group is particularly notable, as sequential analysis indicated the most fragmented game behaviours, with isolated use of the chatbot and the notebook/calculator tool. These students might have prioritised interactions with the chatbot, spending substantial time exploring its chatting features rather than engaging with other game actions or integrating chatbot assistance effectively. As a result, this group had the fewest game attempts among all groups. The novelty effect, combined with the short intervention duration, might have contributed to the chatbot's lack of impact. While this finding may suggest a limitation of GenAI chatbots as standalone scaffolding tools, the novelty effect and limited intervention time could have played a role. Future research should aim to mitigate this novelty effect and extend the intervention period to further explore GenAI's impact on game behaviours.

5.3 | Interaction Between GenAI Chatbot and Instructional Videos in DGBL

While we hoped that the combination of instructional videos and GenAI chatbot might improve students' performance compared to video or chatbot alone, it was not the case in this study. Our finding is similar to Han et al. (2022) but different from Fidan and Gencel (2022) who found that compared with video only, the combination of instructional videos and a chatbot led preservice teachers to better performance. The difference may be attributed to a variety of factors including the difference in learners (preservice teachers vs. secondary students), subject matters (instructional technology vs. science), learning environments (flipped learning vs. DGBL), and types and functionalities of the chatbot (AI chatbot that provided personalised feedback vs. GenAI chatbot that provided a variety of coherent and contextualised support).

Although the combination of videos and the GenAI chatbot did not yield a significant difference in the science learning outcomes of this study, it did decrease the extraneous cognitive load for the video + GenAI group compared with the video only group. We reason that the GenAI chatbot's effectiveness in

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reducing extraneous load stems from the distinct types of scaffolding provided by the chatbot and videos. The instructional videos presented systematic and structured scaffolds on science concepts (e.g., speed, force, distance), yet these supports were detached from the game world (i.e., a village under attack, delivering life supplies). Consequently, students might view these videos as extraneous since they do not directly pertain to the game scenario. In contrast, the chatbot, by responding to students' prompts, provided support that was contextualised within the game world. For those students who had access to both the videos and the chatbot, navigating between the two likely helped them to bridge the decontextualized videos with game elements and actions. For example, when students in the video only group watched a video explaining the concept of inertia, they might not have clearly understood its connection to the game. However, students in the video + AI group were prompted with questions such as, 'When does inertia occur in the game?' The ensuing interactions have helped these students bridge the concept with game mechanics, reducing their perceived extraneous load associated with the video. Our sequential analysis lends some support for this hypothesis in two ways. Firstly, students in the video + GenAI group frequently switched among the chatbot, the game board and the instructional videos, suggesting a certain sense-making process. Secondly, in comparison to the GenAI only group—who interacted with the chatbot in an isolated manner, the video + GenAI group exhibited more integrated game behaviours, indicating the combination of the chatbot and videos prompted a more cohesive learning experience. In conclusion, the instructional videos and the chatbot might have mutually supported and reinforced each other.

While interaction effect was observed on extraneous cognitive load, no such effect was found on intrinsic or germane load. We speculate that the complexity of the physics concepts embedded in the game may have contributed to a high intrinsic load across all four groups. Combined with the short duration of the intervention, this may have led students in all groups to perceive a similar impact on their learning, as reflected in the lack of significant treatment effect on germane load. Taken together, the absence of treatment effects on intrinsic and germane cognitive load, along with the small effect size for the interaction effect on extraneous cognitive load, may explain the lack of an interaction effect on science learning outcomes. More research is needed to replicate our findings and investigate whether a longer intervention period might produce a more substantial effect on cognitive load and potentially improve learning outcomes.

6 | Implications and Conclusions

This study contributes to the existing literature on the utility of instructional videos as a scaffolding mechanism within the DGBL context, highlighting their effectiveness. Additionally, the study is one of the first attempts to explore the use of GenAI chatbots in supporting science learning in DGBL. Although significant impacts of the GenAI chatbot were not observed, our findings point to a few promising directions for future research on harnessing GenAI to support science inquiry. First, SRL skills appear to be crucial for K–12 students to benefit from GenAI-supported chatbots in science inquiries. Future studies should investigate the effects of SRL interventions, either implemented before or during interactions with chatbots (e.g., Chiu, Moorhouse, et al. 2023; Chiu, Xia, et al. 2023; Liang et al. 2024). GenAI enables chatbots to provide adaptive and individualised SRL support, which has shown advantages (Lee et al. 2025). Recent work by Lai (2024) offers a framework for GenAI chatbots to support various SRL processes—such as defining, seeking, engaging and reflecting—providing guidance for chatbot design to support SRL. Secondly, further investigation is needed into how structured, context-independent scaffolds, such as instructional videos, can be effectively integrated with the contextualised and adaptive scaffolds provided by GenAI, particularly for K–12 learners. Third, beyond examining the effect of GenAI on learning outcomes in DGBL, future research should further explore students' in-game behaviours while interacting with GenAI, with a particular focus on potential behavioural fragmentation and strategies to mitigate it. Lastly, while this study identified significant benefits of instructional videos to seventh-grade students, these scaffolds may not be equally effective for all students, as the effects of DGBL can vary across different groups (Cai et al. 2022). Future research should investigate how instructional videos and GenAI chatbots in DGBL influence engagement and science learning outcomes among elementary and college students.

This study has a few limitations. First, the short duration of the treatment may not have been sufficient for students to familiarise themselves with the chatbot and use it effectively. Future research should implement longer interventions and incorporate assessments of knowledge transfer to gain deeper insights into long-term effects. Second, the novelty of the GenAI chatbot might have influenced students' engagement and behaviours, effects that may not be sustained over time. Future studies should incorporate a longer familiarisation period to mitigate this confounding effect. Third, while this study included a control group without the instructional videos and chatbot, a true control group without any scaffolding tools might provide a better baseline understanding of students' performance and behaviours. Fourth, the chatbot examined in this study offered only a text-based modality. Additional modalities, such as images and videos, may significantly enhance its effectiveness, warranting further research. Fifth, a more nuanced behavioural coding scheme could have offered clearer insights into the differences among treatment groups. Sixth, using the same science knowledge test as both pretest and posttest may introduce potential testing effects, though the 2-week interval between the pretest and posttest might minimise this bias. Seventh, surveying students' prior experience with chatbots could provide a more comprehensive understanding of the participants, helping to contextualise the research findings. Lastly, this study focused exclusively on seventh-grade students in Taiwan, which may limit its generalisability to other cultural and educational contexts. Future research should replicate this study across diverse populations.

This study also offers practical strategies for educators and instructional designers seeking to effectively implement scaffolding tools to enhance science learning in DGBL environments. First, educators can utilise instructional videos to deliver just-in-time, systematic information that fosters conceptual understanding. These videos can act as in-game teachers, providing

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timely assistance during gameplay. Secondly, it is important to recognise that integrating GenAI support does not automatically improve learning outcomes. In fact, standalone GenAI chatbots may contribute to fragmented inquiry behaviours in DGBL. Prioritising the training and development of K–12 students' SRL and prompting strategies can help students more effectively leverage GenAI in science inquiries (Lodge et al. 2023; Rejeb et al. 2024; Theophilou et al. 2023). Thirdly, combining a GenAI chatbot with structured knowledge scaffolds, such as instructional videos, may prevent students from engaging in isolated interactions with the chatbot, instead encouraging more strategic and cohesive game behaviours. These strategies can empower teachers and designers to create engaging, effective DGBL experiences that optimise both student learning and motivation.

Author Contributions

Kun Huang: validation, methodology, writing – original draft, resources, writing – review and editing, visualization. **Ching-Huei Chen:** conceptualization, methodology, data curation, supervision, writing – original draft, funding acquisition.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A

Samples of Science Knowledge Test

1. When a coin is thrown vertically upward in an accelerating bus, where will the coin land?

(A) The same spot.

(B) Behind the original spot.

(C) In front of the same spot.

2. When the hanging straps on the bus ceiling swing forward, what does this indicate about the bus's motion?

(A) The bus is decelerating.

(B) The bus is accelerating.

(C) The bus is moving at constant speed.

Appendix B

Example of Context-Sensitive Chatbot Interaction Using System Prompting

Roles	Definition	Excerpt
System	Prompts provided to the chatbot (excerpt)	Learning Objective: • This level focuses on the relationship among 'distance', 'time', and 'speed' in linear motion, using the formula: distance/time = speed. Student Gameplay Status: • Current stage: Basic level • Total incorrect attempts: 2 • Number of speed segments tested using the stopwatch: 2 • Task: Basic objective not yet achieved Based on this information, play the role of a physics teacher who provides guided, step-by-step support (without giving away the answer), using Traditional Chinese in under 100 characters, in a way that a middle school student can understand. Refer to the student as 'you' in responses.
User	The student's question during gameplay	How do I play?
Assistant	The chatbot's response, generated by GPT based on the system prompt and user input	First, choose a speed segment to control the dessert's movement. Then, start the stopwatch and observe the dessert's motion on the ruler platform. You can pause the stopwatch at any time to record the dessert's position and calculate the distance it moved. Then, use the formula distance/time = speed to calculate the speed. Repeat this process until you find the correct speed segment, then check with the Dessert Uncle.

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